

A female powerlifter with blonde hair, wearing a blue and white singlet and a black belt, is celebrating with a male man in a black shirt. They are both smiling and clapping their hands. The background is a blurred indoor setting with a banner that says "Ticket to the IP" and a large "ESPN ELITE SPORTS NUTRITION" logo on the right.

Beyond the Bar: What affects a collegiate powerlifter's bench press weight in competition?

By: Tim & Sophie

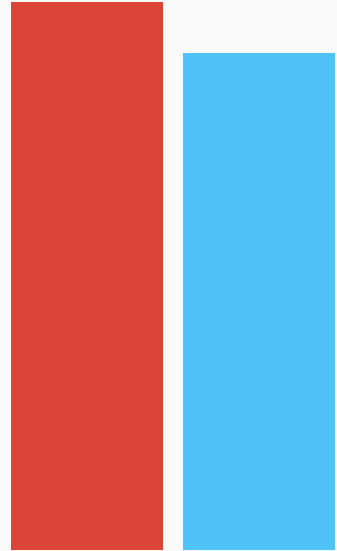
Introduction

Motivation: Curiosity about what influences a heavy and successful bench press weight in collegiate competition.

Research Question:

How can a lifter's bodyweight and sex predict their best bench press weight in competition?

Do age and home-country advantage also play significant roles?



Our Data

Open Powerlifting:

World University Powerlifting
Cup hosted by the International
Powerlifting Federation (IPF).

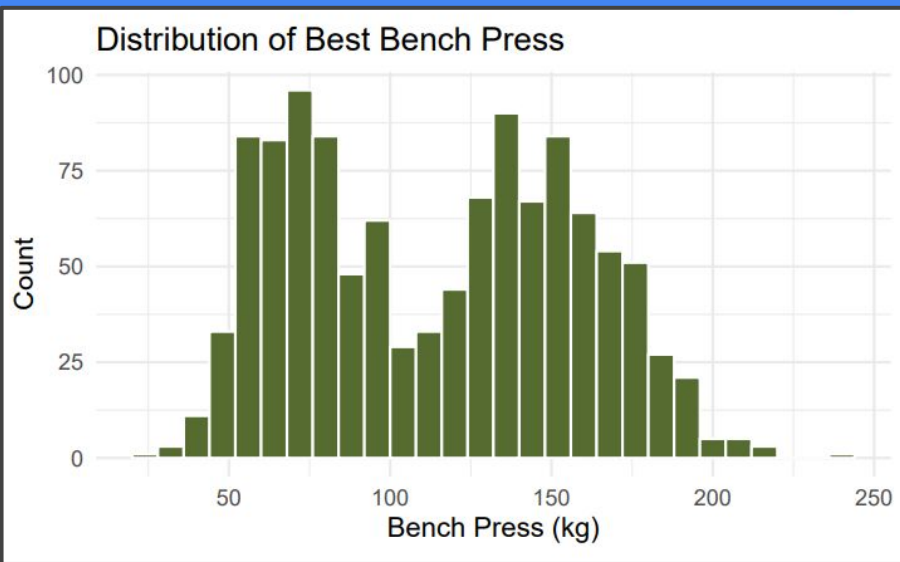
2017-2019, 2021-2025

Sex	Age	BodyweightKg	Best3SquatKg	Best3BenchKg	Best3DeadliftKg	TotalKg	Place
M	16.5	88.25	210.0	120.0	267.5	597.5	18
M	18.0	65.55	150.0	150.0	192.5	492.5	11
M	18.5	65.30	137.5	107.5	185.0	430.0	12
M	19.0	72.95	240.0	147.5	307.5	695.0	3
M	19.0	128.65	217.5	140.0	220.0	577.5	4
M	20.0	81.60	237.5	140.0	250.0	627.5	17
M	20.5	81.40	195.0	140.0	232.5	567.5	21
M	21.0	92.10	230.0	140.0	277.5	647.5	15
M	21.0	102.50	242.5	135.0	247.5	625.0	12
M	21.0	81.90	222.5	132.5	255.0	610.0	18

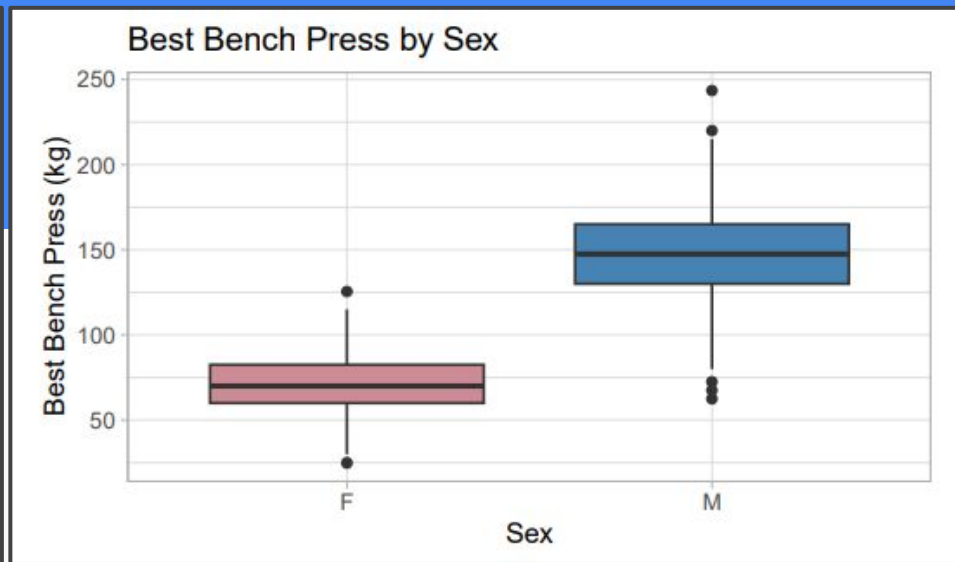
Dots	Country	MeetCountry	Year	CountryEqualMeet
390.24	Turkey	Turkey	2025	1
387.86	Turkey	Turkey	2025	1
339.60	Qatar	Turkey	2025	0
507.67	Turkey	Turkey	2025	1
324.39	Turkey	Turkey	2025	1
427.73	Ireland	Turkey	2025	0
387.38	Bulgaria	Turkey	2025	0
413.92	Greece	Turkey	2025	0
380.68	Greece	Turkey	2025	0
414.93	Greece	Turkey	2025	0

Highlights from Exploratory Data Analysis

Why biological sex must be a predictor:



Two distinct peaks

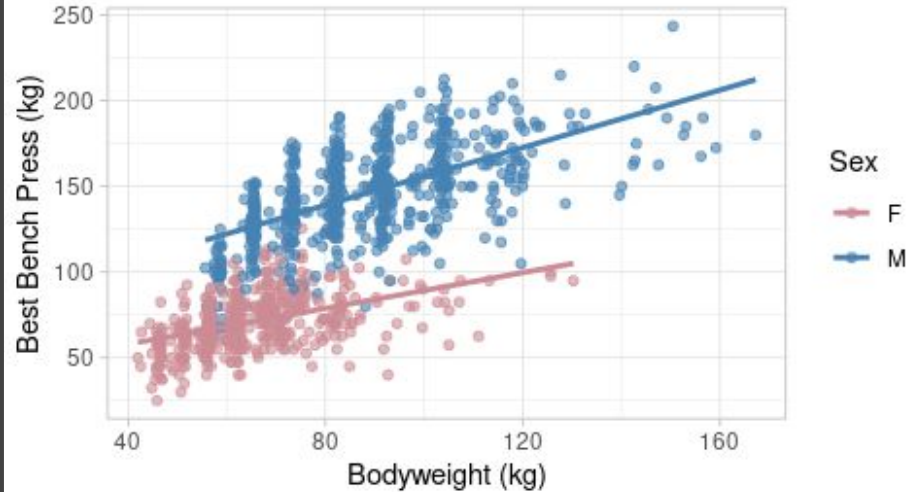


Confirms that the histogram peaks are due to different weights performed by different biological sexes.

Highlights from Exploratory Data Analysis

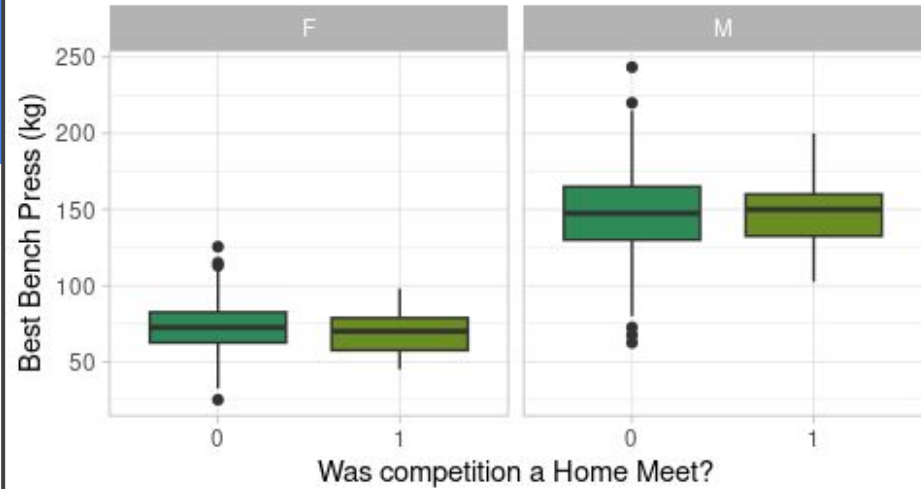
Bodyweight and Home Meet Indicator:

Bodyweight vs Best Bench Press by Sex



Bodyweight is positively correlated with best bench press.

Best Bench Press by Home Meet



Difference between lifters at Home Meets and Away Meets is almost non existent.

Initial Modeling and next steps

term	estimate st
(Intercept)	-6.724
BodyweightKg	0.742
SexM	57.574
Age	1.343

OLS vs GLS

Decide if OLS is sufficient, if not, look into GLS methods

Pick Model

Decide on the best predictors for outcome

Multicollinearity

Discover if there exists high correlation between variables

Test Models

Look at confidence intervals, p-values, R-Squared, AIC, etc.